AP® Statistics Master Review Series - Book 1

AP® Statistics Formula & Inference Decision Guide

The Complete High-Yield Exam Companion: Formula Breakdowns, Mindmaps, Inference Decision Trees, and Rubric Sentence Frames

PR

2027 Exam Ready Edition

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How to Use This Master Companion

Welcome to the **AP Statistics Formula & Inference Decision Guide**! This companion is engineered to replace bulky 600-page textbooks with high-yield clarity, visual mindmaps, and rubric sentence frames.

The 4-Pillar Master System

Pillar 1: **Unit Mindmaps & Flowcharts:** Master the visual architecture of each unit before diving into detailed formulas.

Pillar 2: **Formula & Parameter Translations:** Understand exactly what every Greek symbol ($\mu, \sigma, \hat{p}, \chi^2$) represents in plain English.

Pillar 3: **Mnemonics & Grader Pitfalls:** Use universal mnemonics (`SOCS`, `BINS`, `LINER`, `PANIC`, `PHANTOM`) to avoid losing partial credit.

Pillar 4: **Fill-in-the-Blank Sentence Frames:** Memorize standardized rubric templates to capture 100% credit on Free-Response Questions (FRQs).

Official AP® Statistics Exam Blueprint

1. Exam Format & Pacing Strategy (3 Hours / 180 Mins)

Section	Questions	Time	Weight	Pacing Strategy
Section I: MCQs	40 Questions	90 mins	50%	2.25 mins / question
Section II: FRQs	6 Questions	90 mins	50%	Q1–5: 13 mins each
Part A (Q1–5)	5 Standard	65 mins	37.5%	Core concepts
Part B (Q6)	1 Invest. Task	25 mins	12.5%	Novel applications

2. Official 9-Unit Syllabus Weightage Matrix

Unit	Curriculum Topic	Exam Weight	Yield Priority
Unit 1	Exploring One-Variable Data	15–23%	**Tier 1 (High)**
Unit 2	Exploring Two-Variable Data	5–7%	**Tier 2 (Medium)**
Unit 3	Collecting Data	12–15%	**Tier 1 (High)**
Unit 4	Probability, Random Variables & Dist.	10–20%	**Tier 1 (High)**
Unit 5	Sampling Distributions	7–12%	**Tier 2 (Medium)**
Unit 6	Inference for Proportions (Categorical)	12–15%	**Tier 1 (High)**
Unit 7	Inference for Means (Quantitative)	10–18%	**Tier 1 (High)**
Unit 8	Chi-Square Tests	2–5%	**Tier 3 (Quick)**
Unit 9	Inference for Slopes	2–5%	**Tier 3 (Quick)**

Exam-Day Morning Routine & Master Mnemonics

Exam-Day Morning Checklist

- **TI-84 Plus CE Inspection:** Charge your calculator to 100% the night before. Ensure RAM is clean and degree/radian mode does not interfere with list stats.
- **Digital Bluebook Readiness:** Verify your device is fully charged, software is updated, and your College Board login credentials are confirmed.
- **The 5-Minute Brain Dump:** When the exam starts, write down key mnemonics (`SOCS`, `BINS`, `LINER`, `PANIC`, `PHANTOM`) on your scratch paper.

The Top 5 Universal AP Statistics Mnemonics

1. SOCS — Describing Quantitative Distributions (Unit 1)

S — Shape: Skewed Left, Skewed Right, Symmetric, Unimodal, Bimodal.
O — Outliers: Apply $1.5 \times$ IQR rule or 2SD rule.
C — Center: Report Median (for skewed) or Mean (for symmetric).
S — Spread: Report IQR/Range (for skewed) or Standard Deviation (for symmetric).
Always include context and units!

2. BINS — Binomial Distribution Checks (Unit 4)

B — Binary: Only two possible outcomes (Success vs Failure).
I — Independent: Trials must be independent (or 10% condition holds).
N — Number: Fixed number of trials (n).
S — Same Probability: Probability of success (p) remains constant.

3. LINER — Linear Regression Inference Conditions (Unit 9)

L — Linear: Scatterplot shows linear pattern; residual plot shows random scatter.

I — Independent: 10% condition ($n \leq 0.10N$).

N — Normal Residuals: Histogram/normal probability plot of residuals is roughly normal.

E — Equal Variance: Residuals show equal spread across all $\hat{y}$ values.

R — Random: Data collected via random sampling or randomized experiment.

4. PANIC — 5-Step Confidence Interval Protocol

P — Parameter of interest defined in context (μ or p).

A — Assumptions & Conditions checked (Random, 10%, Normal / Large Counts).

N — Name the specific interval procedure (e.g., 1-Sample t-Interval for μ).

I — Interval calculated with formula and calculator degrees of freedom.

C — Conclusion stated using standard rubric sentence frames in context.

5. PHANTOM — 7-Step Hypothesis Testing Protocol

P — Parameter defined in context.

H — Hypotheses stated (H_0 and H_a using parameter symbols).

A — Assumptions checked.

N — Name the test procedure (e.g., 2-Prop z-Test).

T — Test statistic calculated (z, t, or χ^2).

O — Obtain p-value and report degrees of freedom.

M — Make decision: Reject H_0 or Fail to Reject H_0 with contextual conclusion.

Contents

Chapter 1

Unit 1: Exploring One-Variable Data

1.1 Unit 1 Visual Mindmap

Unit 1: One-Variable Data (15–23%)

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Data Types: Categorical vs Quantitative

↓

Description: SOCS (Shape, Outliers, Center, Spread)

↓

Outliers: Lower $= Q_1 - 1.5\text{IQR}$, Upper $= Q_3 + 1.5\text{IQR}$

↓

Normal Model: $z = \frac{x-\mu}{\sigma}$, 68-95-99.7 Rule

1.2 Essential Formulas & Plain-English Translation

Unit 1 Core Formulas

1. Sample Mean ($\bar{x}$): $\bar{x} = \frac{\sum x_i}{n}$ — The arithmetic average; non-resistant to skew/outliers.

2. Sample Standard Deviation (s_x): $s_x = \sqrt{\frac{\sum(x_i-\bar{x})^2}{n-1}}$ — The typical distance of observations from the mean.

3. Standardized Score (z-score): $z = \frac{x-\mu}{\sigma}$ — Number of standard deviations an observation falls above (+) or below (−) the mean.

1.3 Resistant vs. Non-Resistant Summary Statistics

Measure	**Resistant to Outliers / Skew?**	**When to Use on Exam**
Median & IQR	**Yes** (Middle 50% only)	Skewed data or data with outliers
Mean & Standard Deviation	**No** (Pulled toward the tail/extremes)	Symmetric, bell-shaped distributions

1.4 Top Exam Traps & Sentence Frames

Unit 1 Trap Alert: Comparative Language

When comparing two distributions on an FRQ, you **must** use comparative words (*"greater than", "less than"*). Stating two separate descriptions scores zero credit!

Unit 1 FRQ Sentence Frame: z-Score Interpretation

"The observed value of [context] is [z-score value] standard deviations [above/below] the mean of [context with units]."

Chapter 2

Unit 2: Exploring Two-Variable Data

2.1 Unit 2 Visual Mindmap

Unit 2: Two-Variable Data (5–7%)

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Scatterplots: Direction (+/−), Form, Strength, Outliers

↓

LSRL Line: $\hat{y} = a + bx$ ($\hat{y}$ = predicted response)

↓

Residuals: Residual $= y - \hat{y}$ (Actual minus Predicted)

↓

Coefficients: Correlation r, Determination r^2, SE s

2.2 Essential Formulas & Parameter Breakdowns

Unit 2 Core Formulas: Linear Regression

1. **Least-Squares Regression Line:** $\hat{y} = a + bx$ ($\hat{y}$ = predicted response)
2. **Slope (b):** $b = r\left(\frac{s_y}{s_x}\right)$ — Change in predicted y per 1-unit increase in x.
3. **y-Intercept (a):** $a = \bar{y} - b\bar{x}$ — The point $(\bar{x}, \bar{y})$ is always on the LSRL.
4. **Residual (e):** Residual $= y - \hat{y} =$ Actual y − Predicted $\hat{y}$

2.3 Sentence Templates for FRQs

Unit 2 FRQ Sentence Templates

1. Slope (b): *"For each additional 1 [unit of x], the predicted [response variable y] increases/decreases by approximately [value of b] [units of y]."*
2. r^2: *"Approximately [value of $r^2 \times 100\%$] of the variation in [response variable in context] is accounted for by the linear model with [explanatory variable in context]."*
3. Standard Error of Residuals (s): *"The typical distance between the actual [response variable y] and the predicted values from the regression line is approximately [s units of y]."*

Chapter 3

Unit 3: Collecting Data

3.1 Unit 3 Visual Flowchart: Sampling vs. Experimentation

Unit 3: Collecting Data (12–15%)

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Observational Studies: Shows association only.

↓

Experiments: Random assignment establishes **Causation**.

↓

Sampling Methods: SRS, Stratified, Cluster, Systematic.

↓

Bias Types: Undercoverage, Nonresponse, Response.

3.2 The 4 Principles of Experimental Design

1. **Comparison:** At least two treatment groups (or treatment vs. control) to isolate effects.
2. **Random Assignment:** Balances the effects of confounding variables across treatment groups.
3. **Control:** Keep all other outside variables constant.
4. **Replication:** Apply treatments to sufficiently many experimental units.

Unit 3 Trap Alert: Blocking vs. Stratification

Stratification is used when *sampling* from a population.

Blocking is used when *assigning treatments* in an experiment.

Never mix these terms on an AP exam!

Chapter 4

Unit 4: Probability, Random Variables, & Distributions

4.1 Essential Formulas & Probability Rules

Unit 4 Core Formulas

1. Addition Rule: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ (If disjoint, $P(A \cap B) = 0$).
2. Conditional Probability: $P(A \mid B) = \frac{P(A \cap B)}{P(B)}$
3. Independence Test: Events A and B are independent if $P(A \mid B) = P(A)$ or $P(A \cap B) = P(A) \cdot P(B)$.
4. Combining Random Variables:

$$\mu_{X \pm Y} = \mu_X \pm \mu_Y$$

$$\sigma^2_{X \pm Y} = \sigma^2_X + \sigma^2_Y \quad (\textbf{Always ADD variances!})$$

5. Binomial Distribution $B(n, p)$:

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

$$\mu_X = np, \qquad \sigma_X = \sqrt{np(1-p)}$$

Unit 4 Trap Alert: Independence vs Disjoint

Mutually exclusive (disjoint) events can **NEVER** be independent (if $P > 0$). Knowing A happened means B cannot happen!

Chapter 5

Unit 5: Sampling Distributions

5.1 Unit 5 Core Formulas & Central Limit Theorem

Sampling Distributions

1. Sample Proportion ($\hat{p}$):

$$\mu_{\hat{p}} = p, \qquad \sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$$

Normal if $np \geq 10$ and $n(1-p) \geq 10$.

2. Sample Mean ($\bar{x}$):

$$\mu_{\bar{x}} = \mu, \qquad \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

Normal if $n \geq 30$ by CLT or population is normal.

Central Limit Theorem (CLT) Golden Rule

The CLT states that the **sampling distribution of $\bar{x}$** becomes approximately normal as n increases ($n \geq 30$), regardless of the shape of the underlying population! It does *not* apply to individual observations.

Chapter 6

Unit 6: Inference for Proportions (Categorical Data)

6.1 1-Proportion vs. 2-Proportion Inference Procedures

Unit 6 Formulas

1-Prop z-Interval: $\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$
1-Prop z-Test: $z = \frac{\hat{p}-p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$ (Use p_0 for Large Counts condition!)
2-Prop z-Test (Pooled):

$$\hat{p}_c = \frac{x_1 + x_2}{n_1 + n_2}$$

$$z = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{\sqrt{\hat{p}_c(1-\hat{p}_c)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

Unit 6 PANIC Template for 1-Prop z-Interval

P: Let $p =$ true proportion of [context].
A: Random sample; 10% condition ($n \leq 0.10N$); Large counts ($n\hat{p} \geq 10, n(1-\hat{p}) \geq 10$).
N: 1-Proportion z-Interval.
I: Record $[\text{Lower}, \text{Upper}]$ with 95% confidence.
C: *"We are 95% confident that the interval from [Lower] to [Upper] captures the true proportion of [context]."*

Chapter 7

Unit 7: Inference for Means (Quantitative Data)

7.1 1-Sample t, Paired t, and 2-Sample t Inference

Unit 7 Means Formulas

1-Sample t-Interval: $\bar{x} \pm t^* \left(\frac{s}{\sqrt{n}}\right) \quad (\text{df} = n - 1)$
1-Sample t-Test: $t = \frac{\bar{x}-\mu_0}{s/\sqrt{n}} \quad (\text{df} = n - 1)$
2-Sample t-Test: $t = \frac{(\bar{x}_1-\bar{x}_2)-0}{\sqrt{\frac{s_1^2}{n_1}+\frac{s_2^2}{n_2}}}$ (`Pooled: NO`)

Unit 7 Trap Alert: Paired t-Test vs. 2-Sample t-Test

If data comes from pre/post tests, twins, or the same experimental unit measured twice, it is a **Paired t-Test** on the single list of differences $(L_1 - L_2)$. Never run a 2-sample t-test on paired data!

Chapter 8

Unit 8: Chi-Square Tests

8.1 The 3 Types of Chi-Square Tests

Test Name	Number of Samples	Variables	Degrees of Freedom
χ^2 **Goodness of Fit**	1 Sample	1 Categorical Variable	$\text{df} = k - 1$
χ^2 **Homogeneity**	2+ Independent Samples	1 Categorical Variable	$\text{df} = (r-1)(c-1)$
χ^2 **Independence**	1 Single Sample	2 Categorical Variables	$\text{df} = (r-1)(c-1)$

Chi-Square Test Statistic

$$\chi^2 = \sum \frac{(\text{Observed} - \text{Expected})^2}{\text{Expected}}$$

$$\text{Expected Count} = \frac{(\text{Row Total}) \times (\text{Column Total})}{\text{Table Total}}$$

Unit 8 Condition Alert

All **Expected Counts** must be ≥ 5. Never check observed counts for this condition!

Chapter 9

Unit 9: Inference for Slopes

9.1 Linear Regression Slope t-Test & t-Interval

Unit 9 Formulas

t-Interval for Slope β: $b \pm t^* \cdot \mathrm{SE}_b \quad (\mathrm{df} = n - 2)$

t-Test for Slope β: $t = \frac{b-0}{\mathrm{SE}_b} \quad (\mathrm{df} = n - 2)$

LINER Condition Checklist for Slopes

L — Linear pattern in scatterplot.

I — Independent observations (10% rule).

N — Normal distribution of residuals.

E — Equal variance of residuals across line.

R — Random sampling or assignment.

Chapter 10

Master Inference Selection Decision Flowchart

Data Type	Groups	Goal	Procedure Name	TI-84 Path
Categorical	1 Sample	Estimate p	**1-Prop z-Interval**	`STAT` → `TESTS` → `A`
Categorical	1 Sample	Test p	**1-Prop z-Test**	`STAT` → `TESTS` → `5`
Categorical	2 Samples	Estimate $p_1 - p_2$	**2-Prop z-Interval**	`STAT` → `TESTS` → `B`
Categorical	2 Samples	Test $p_1 = p_2$	**2-Prop z-Test (Pooled)**	`STAT` → `TESTS` → `6`
Categorical	1 Sample (Fit)	Fit dist.	**χ^2 GOF-Test**	`STAT` → `TESTS` → `D`
Categorical	2+ Samples	Homogeneity	**χ^2 Test of Homogeneity**	`STAT` → `TESTS` → `C`
Categorical	1 Sample (2 Var)	Association	**χ^2 Test of Independence**	`STAT` → `TESTS` → `C`
Quantitative	1 Sample	Estimate μ	**1-Sample t-Interval**	`STAT` → `TESTS` → `8`
Quantitative	1 Sample	Test μ	**1-Sample t-Test**	`STAT` → `TESTS` → `2`
Quantitative	Matched Pairs	Difference μ_d	**Paired t-Test/Interval**	`T-Test on` $L_1 - L_2$
Quantitative	2 Indep Samples	Compare $\mu_1 - \mu_2$	**2-Sample t-**	`STAT` → `TESTS` → `4/0`

Chapter 11

Top 25 "Never-Forget" AP Grader Deduction Traps

1. **NEVER say "Accept H_0":** Always write *"Fail to reject H_0"*.
2. **NEVER write "We prove H_a":** Write *"We have convincing statistical evidence for H_a"*.
3. **Hypotheses in parameters (μ, p, β), NEVER statistics ($\bar{x}, \hat{p}, b$).**
4. **Always define parameters in context with units.**
5. **Comparative language required on all distribution comparison FRQs (greater, lesser).**
6. **Always put a hat ($\hat{y}$) on regression predicted values.**
7. **Correlation (r) measures LINEAR strength only (never use r for curved data).**
8. **Association is NOT Causation.**
9. **Blocking is for experiments; Stratification is for sampling.**
10. **Mutually exclusive events are NEVER independent.**
11. **Always ADD variances (σ^2), NEVER add standard deviations (σ).**
12. **CLT applies to the sampling distribution of $\bar{x}$, not sample data.**
13. **Use hypothesized p_0 for 1-Prop z-Test Large Counts ($np_0 \geq 10, n(1-p_0) \geq 10$).**
14. **Use sample $\hat{p}$ for 1-Prop z-Interval Large Counts ($n\hat{p} \geq 10, n(1-\hat{p}) \geq 10$).**
15. **Chi-square conditions require EXPECTED counts ≥ 5, not observed counts.**

16. **Chi-square df:** df = (rows − 1)(cols − 1).
17. **Never pool standard deviations for 2-sample t-tests (`Pooled: NO`).**
18. **Confidence Interval (plausible values) vs Confidence Level (long-run capture rate).**
19. **A high r^2 does NOT prove linearity (always check residual plot for random scatter).**
20. **Voluntary response produces severe bias; results cannot be generalized.**
21. **State test conclusion in terms of H_a in context.**
22. **State the significance level (α) explicitly in decision sentences.**
23. **Calculator Speak alone scores ZERO on FRQs (must write formulas and parameter names).**
24. **Extrapolation is dangerous (do not predict $\hat{y}$ outside the domain of x).**
25. **Context, Context, Context: Include problem units and scenario in every sentence!**

Chapter 12

Standard FRQ Interpretation Sentence Frames

Frame 1: Confidence Interval for a Parameter

"We are [C%] confident that the interval from [Lower] to [Upper] captures the true [proportion / mean] of [context with units]."

Frame 2: Confidence Level ($C\%$)

"If we take many random samples of the same size and construct [C%] confidence intervals, about [C%] of the intervals will capture the true [parameter in context]."

Frame 3: p-Value in Context

"Assuming that [null hypothesis H_0 in context] is true, there is a [p-value] probability of obtaining a sample statistic as extreme as or more extreme than observed, purely by random chance."

Frame 4: Reject H_0 Conclusion

"Because the p-value ([p]) is less than α ([0.05]), we reject H_0. There is convincing statistical evidence that [state H_a in context]."

Frame 5: Fail to Reject H_0 Conclusion

"Because the p-value ([p]) is greater than α ([0.05]), we fail to reject H_0. There is NOT convincing statistical evidence that [state H_a in context]."